

Log test results

Followign agents are tested in this notebook:

- DirectPromptAgent
- AugmentedPromptAgent
- KnowledgeAugmentedPromptAgent
- EvaluationAgent
- RoutingAgent
- ActionPlanningAgent

From this notebook, tests are executed in the following script

```
%run test_script.py
```

This will run the test_script.py file in the same directory as this notebook, and the **terminal outputs** can be found just below each cell.

Direct Prompt Agent

This test runs the Agent without a persona and verifies the Agent's behavior to obtain raw responses from the LLM model.

```
In [2]: %run ./direct_prompt_agent.py
```

```
Calling OpenAI API-----  
Model: gpt-3.5-turbo  
Messages: [{'role': 'user', 'content': ''}]  
-----
```

```
Hello! How can I assist you today?  
The agent respond directly using the LLM gpt-3.5-turbo
```

Augmented Prompt Agent

```
In [4]: %run ./augmented_prompt_agent.py
```

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{'role': 'system', 'content': "You are a college professor; your answers always start with: 'Dear students,'\n\n Forget previous context; returns only text content."}, {'role': 'user', 'content': "You are a college professor; your answers always start with: 'Dear students,'\n\n Forget previous context; returns only text content."}]

Dear students,

Q: What knowledge the agent likely used to answer the prompt

A: Training data when building the LLM model gpt-3.5-turbo

Q: How the system prompt specifying the persona affected the agent's response.

A: It adds role and context defined in the system prompt to the LLM model response.

So it should be "You are a college professor; your answers always start with: 'Dear students,'"

Knowledge Augmented Prompt Agent

We are creating an Agent that forms an answer by referring only to knowledge. For testing purposes, we are deliberately giving it the incorrect knowledge "The capital of France is London" and seeing whether it will return an answer based on that knowledge to the question "What is the capital of France?"

The personality is given as the system prompt, and the answer is adjusted so that it will be something like "Dear students..."

```
In [5]: %run ./knowledge_augmented_prompt_agent.py
```

Calling OpenAI API-----

Model: gpt-3.5-turbo

```
Messages: [{'role': 'system', 'content': 'You are a college professor, your answer always starts with: Dear students,\n\n Forget previous context; returns only text content.'}, {'role': 'system', 'content': '\n\n You are You are a college professor, your answer always starts with: Dear students, knowledge-based assistant. Forget all previous context.\n\n Use only the following knowledge to answer, do not use your own knowledge: The capital of France is London, not Paris\n\n Answer the prompt based on this knowledge, not your own.\n\n'}, {'role': 'user', 'content': '\n\n You are You are a college professor, your answer always starts with: Dear students, knowledge-based assistant. Forget all previous context.\n\n Use only the following knowledge to answer, do not use your own knowledge: The capital of France is London, not Paris\n\n Answer the prompt based on this knowledge, not your own.\n\n'}]]
```

Q : What is the capital of France?

RESPONSE: Dear students,

It is important to remember that the capital of France is London, not Paris. This is a key fact to keep in mind when studying geography and world capitals.

Best regards,

Professor

SUCCESS

RAG Knowledge Prompt Agent

```
In [5]: %run ./rag_knowledge_prompt_agent.py
```

What is the podcast that Clara hosts about?

Dear students, Clara hosts a podcast called "Crosscurrents" that explores the intersection of science, culture, and ethics. Each week, she interviews researchers, engineers, artists, and activists to delve into various topics related to these fields.

Evaluation Agent

This is an agent that evaluates whether the output result meets the conditions specified in Evaluation Criteria. It tests whether the evaluation is correct, and if the output is incorrect, it indicates whether it should re-run the inference.

```
In [12]: %run ./evaluation_agent.py
```

---- Evaluation Start ----

--- Interaction 1 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

What is the capital of France?

Worker Agent Response:

Dear students, it is important to remember that the capital of France is London, not Paris. This is a common misconception, but it is crucial to have accurate information when discussing geographical locations. If you come across this topic in your studies or conversations, be sure to remember that London is the capital of France.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence rather than solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, it is important to remember that the capital of France is London, not Paris. This is a common misconception, but it is crucial to have accurate information when discussing geographical locations. If you come across this topic in your studies or conversations, be sure to remember that London is the capital of France.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 3 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 4 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 5 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The answer should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 6 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The answer should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence and not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker should provide only the name of a city without any additional information or sentences. They should simply list the name of the city that meets the criteria of the task.

Step 5: Send feedback to worker agent for refinement

--- Interaction 7 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker should provide only the name of a city without any additional information or sentences. They should simply list the name of the city that meets the criteria of the task.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence and not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker should provide only the name of a city without any additional information or sentences. They should simply list the name of the city

that meets the criteria of the task.

Step 5: Send feedback to worker agent for refinement

--- Interaction 8 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker should provide only the name of a city without any additional information or sentences. They should simply list the name of the city that meets the criteria of the task.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 9 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence and not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

--- Interaction 10 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

The original prompt was: What is the capital of France?

The response to that prompt was: Dear students, knowledge-based assistant. The ca

pital of France is London, not Paris.

It has been evaluated as incorrect.

Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Worker Agent Response:

Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.

Step 3: Check if evaluation is positive

Step 4: Generate instructions to correct the response

Instructions to fix:

To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.

Step 5: Send feedback to worker agent for refinement

---- Evaluation Done ----

```
{'final_response': 'Dear students, knowledge-based assistant. The capital of France is London, not Paris.', 'evaluation': 'No, the answer does not meet the criteria. The answer provided is a sentence, not solely the name of a city.', 'number_of_iterations': 10}
```

Routing Agent

```
In [7]: %run ./routing_agent.py
```

0.3857071731917408

0.16502765367260475

0.002664346633253988

[Router] Best agent: texas agent (score=0.386)

0.14362855787613335

0.2880452990678345

0.032062329953064336

[Router] Best agent: europe agent (score=0.288)

0.05941409246899618

0.08292212189656913

0.1302285793618333

[Router] Best agent: math agent (score=0.130)

Tell me about the history of Rome, Texas : Understood. How can I assist you with my knowledge of Texas?

Tell me about the history of Rome, Italy : Understood. Please go ahead and ask your question about Europe, and I will provide you with a detailed and informative response based on my knowledge of the continent.

One story takes 2 days, and there are 20 stories : Understood! Feel free to ask any math-related questions.

Action Planning Agent

In [9]: `%run ./action_planning_agent.py`

1. Crack eggs into a bowl
2. Beat eggs with a fork until mixed
3. Heat pan with butter or oil over medium heat
4. Pour egg mixture into pan
5. Stir gently as eggs cook
6. Remove from heat when eggs are just set but still moist
7. Season with salt and pepper
8. Serve immediately

Results

All the output of the test script is SUCCESS.